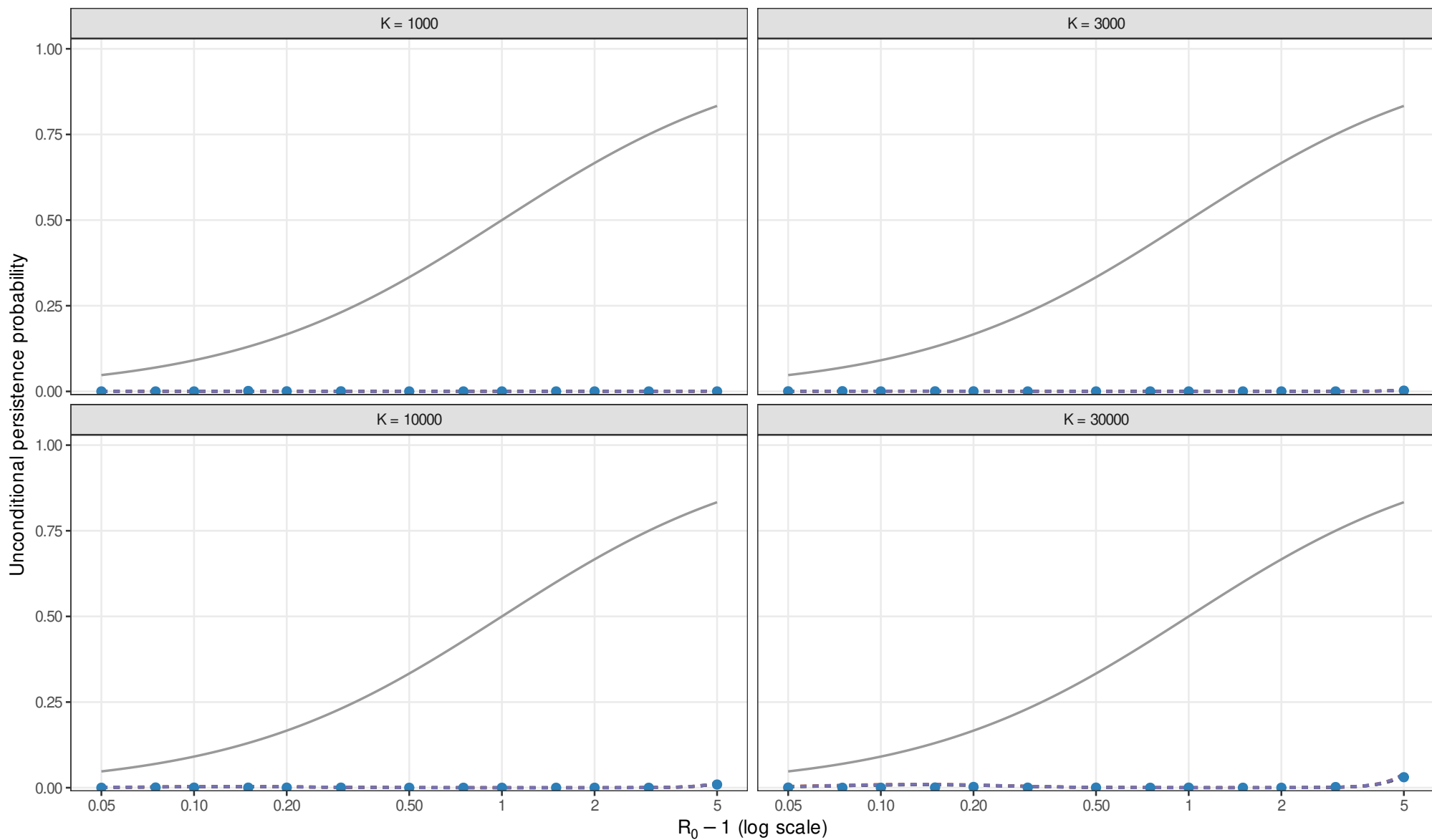


Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.01$, $\theta = 0$; $l_0 = 1$; matching height = y^*



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

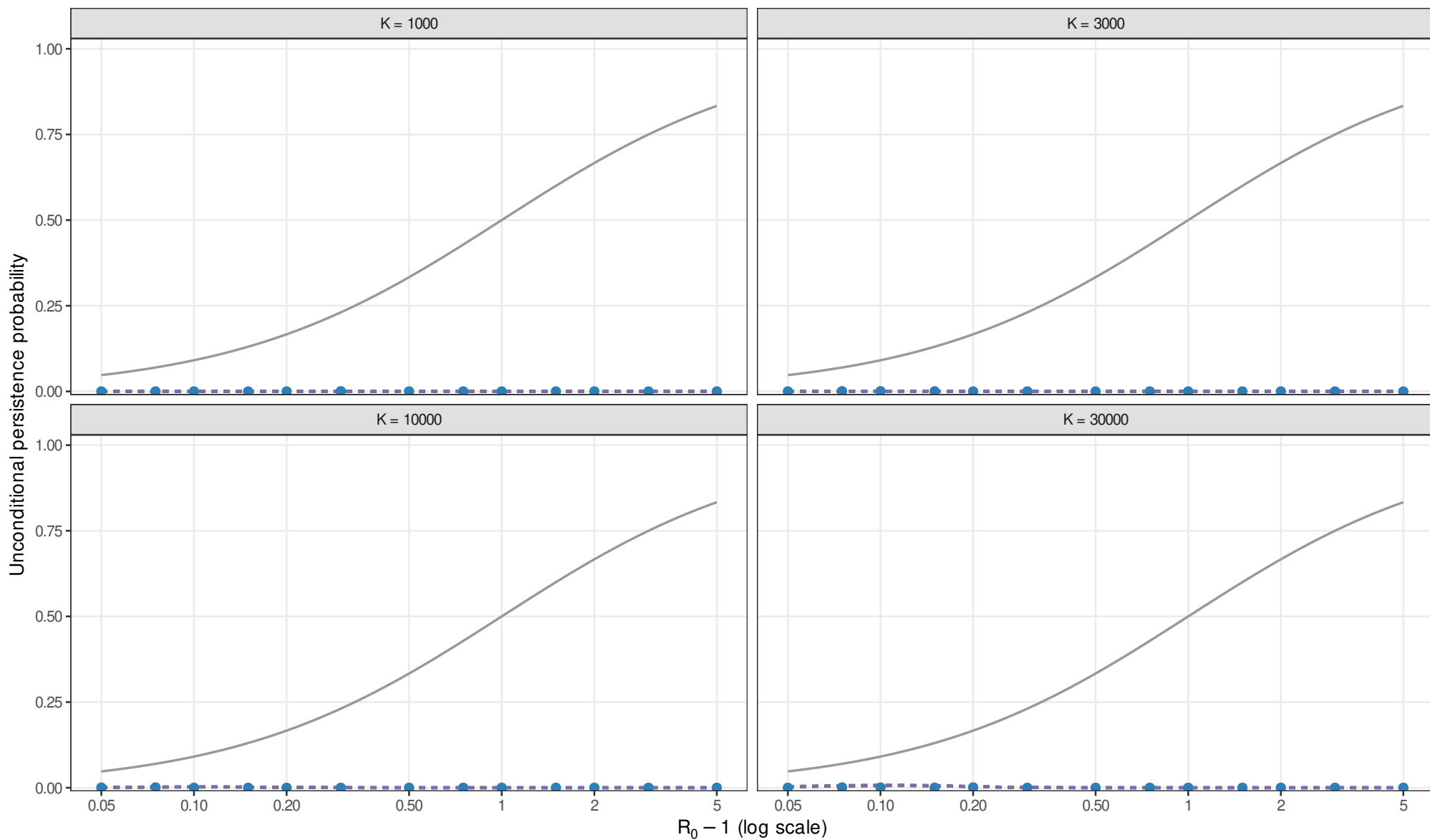
Matching height

- $y_{BL} = y_{star}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height y^* . Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.01$, $\theta = 0.5$; $I_0 = 1$; matching height = y^*



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

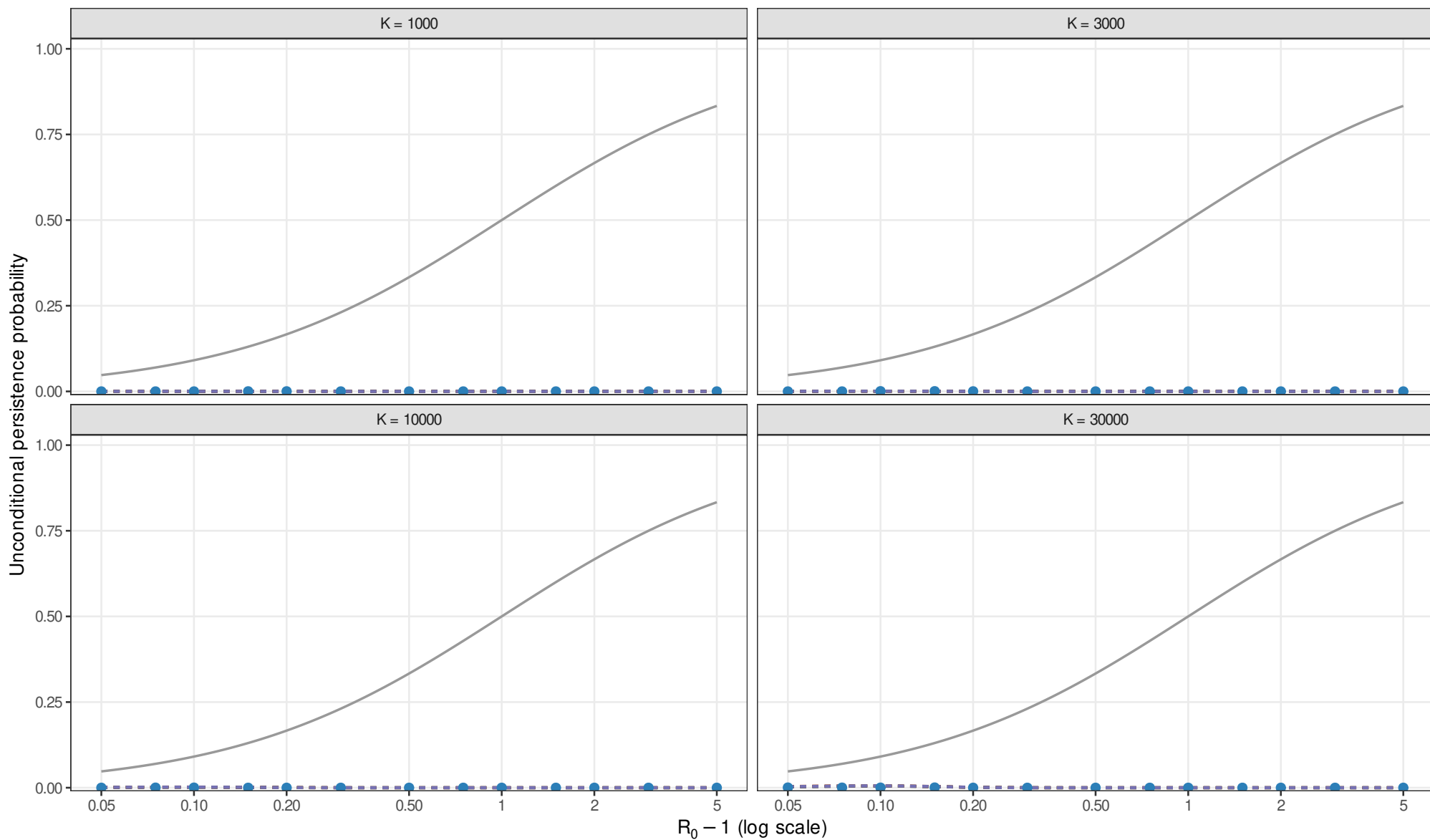
Matching height

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Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.01$, $\theta = 1$; $l_0 = 1$; matching height = y^*



Analytical method

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- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

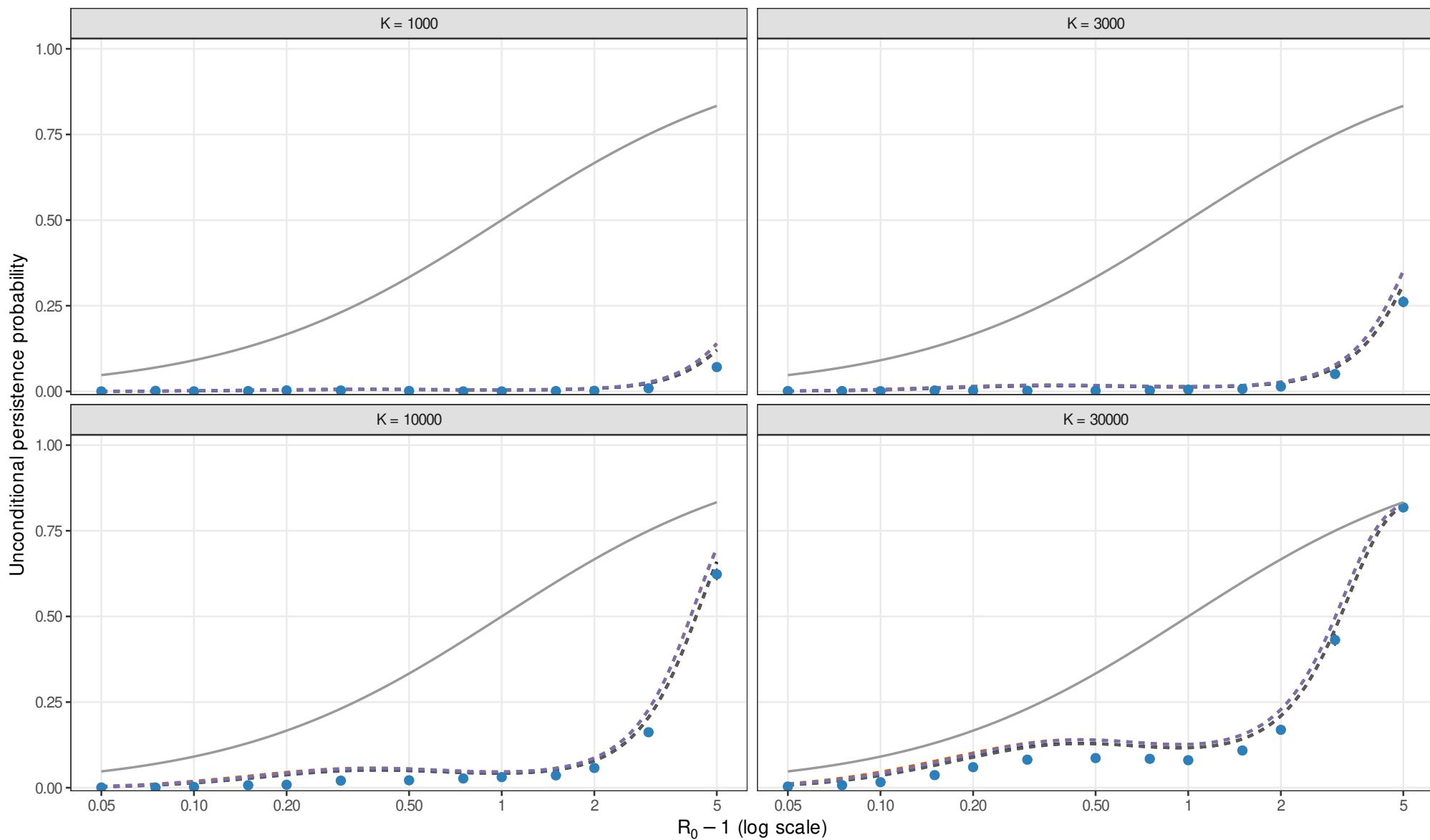
Matching height

- $y_{BL} = y_{star}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height y^* .
Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.02$, $\theta = 0$; $l_0 = 1$; matching height = y^*



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

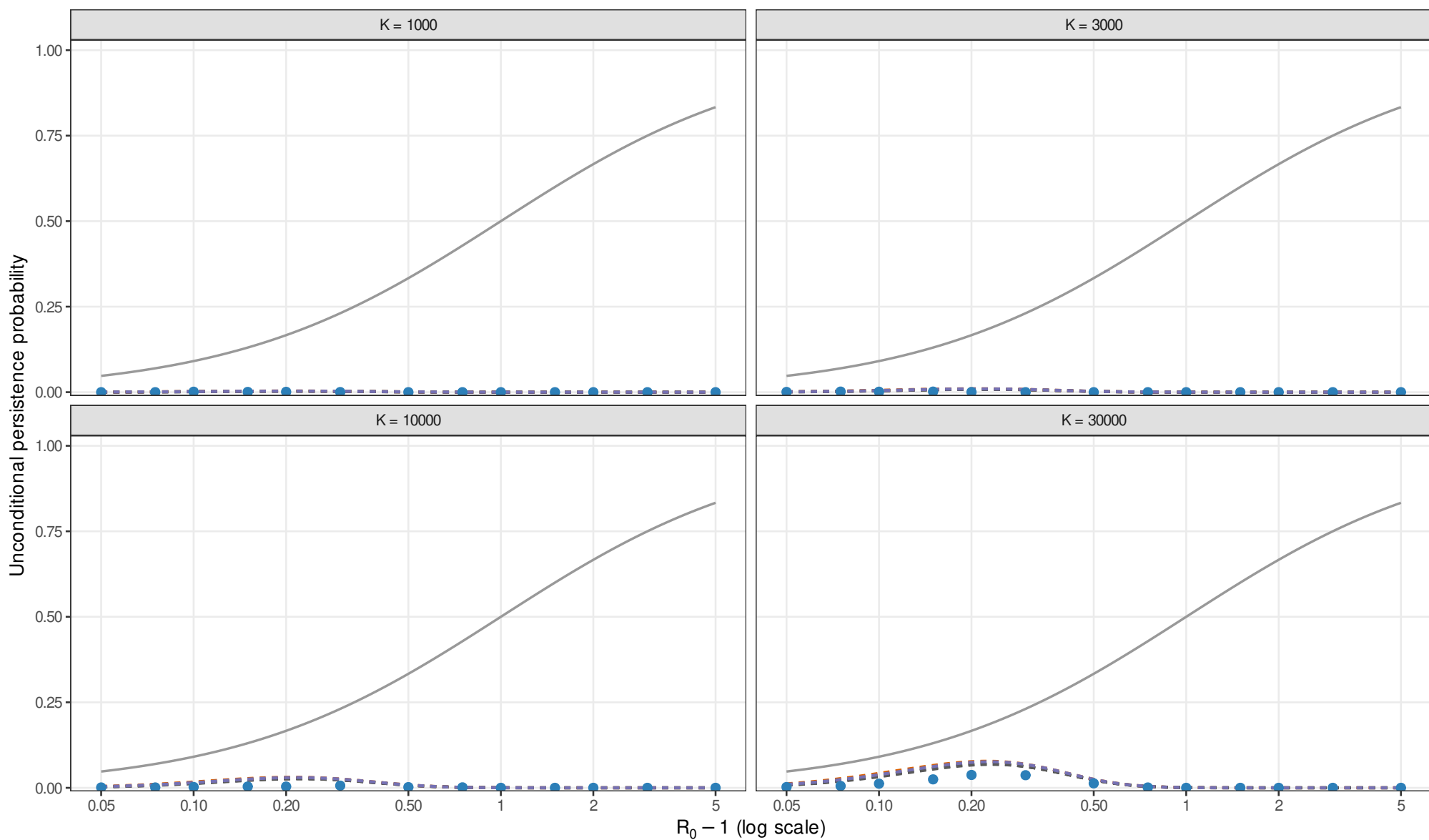
Matching height

- $y_{BL} = y_{star}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height y^* . Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.02$, $\theta = 0.5$; $I_0 = 1$; matching height = y^*



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

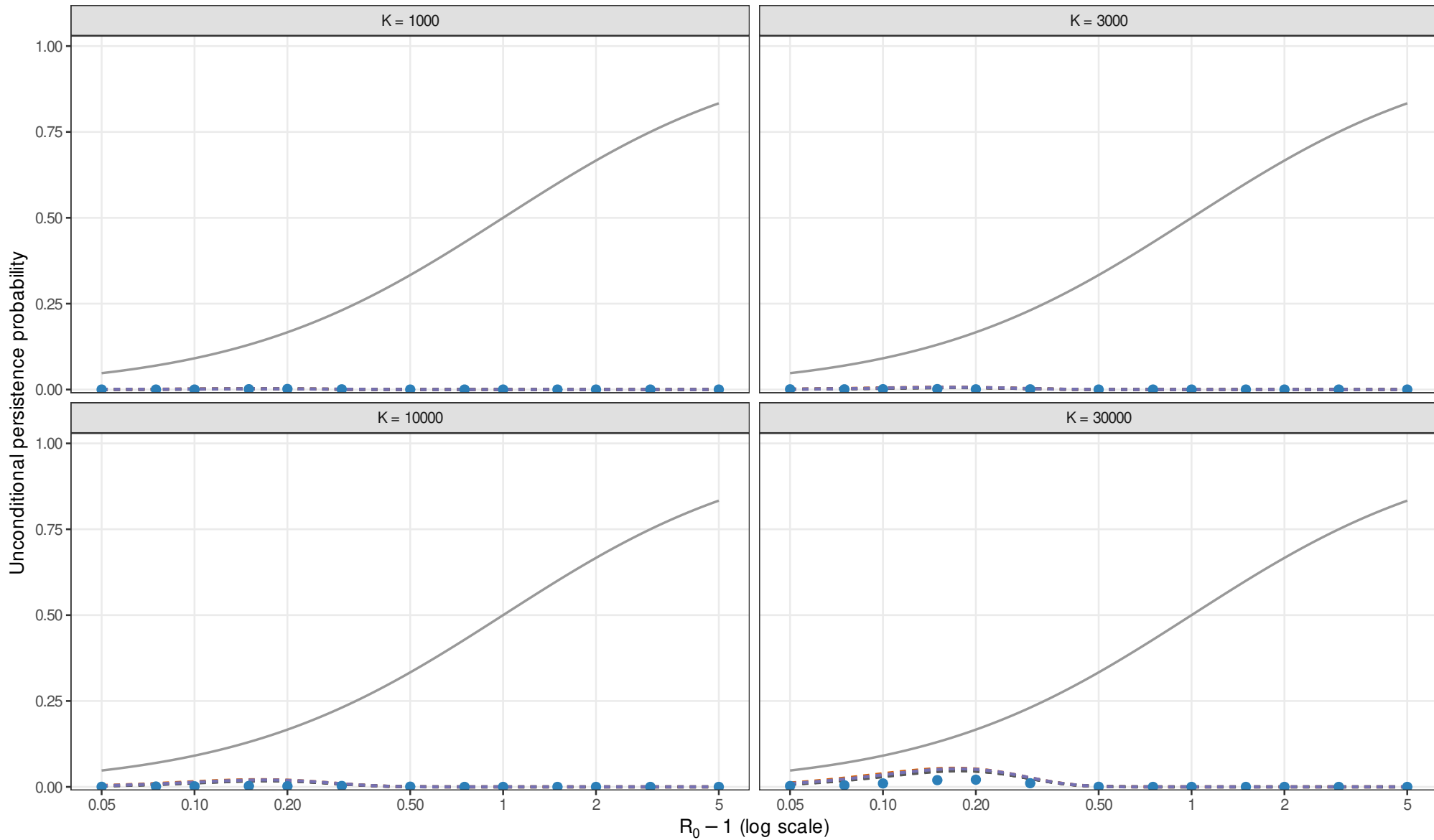
Matching height

- $y_{BL} = y_{star}$
- Early-establishment reference

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Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.02$, $\theta = 1$; $l_0 = 1$; matching height = y^*



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

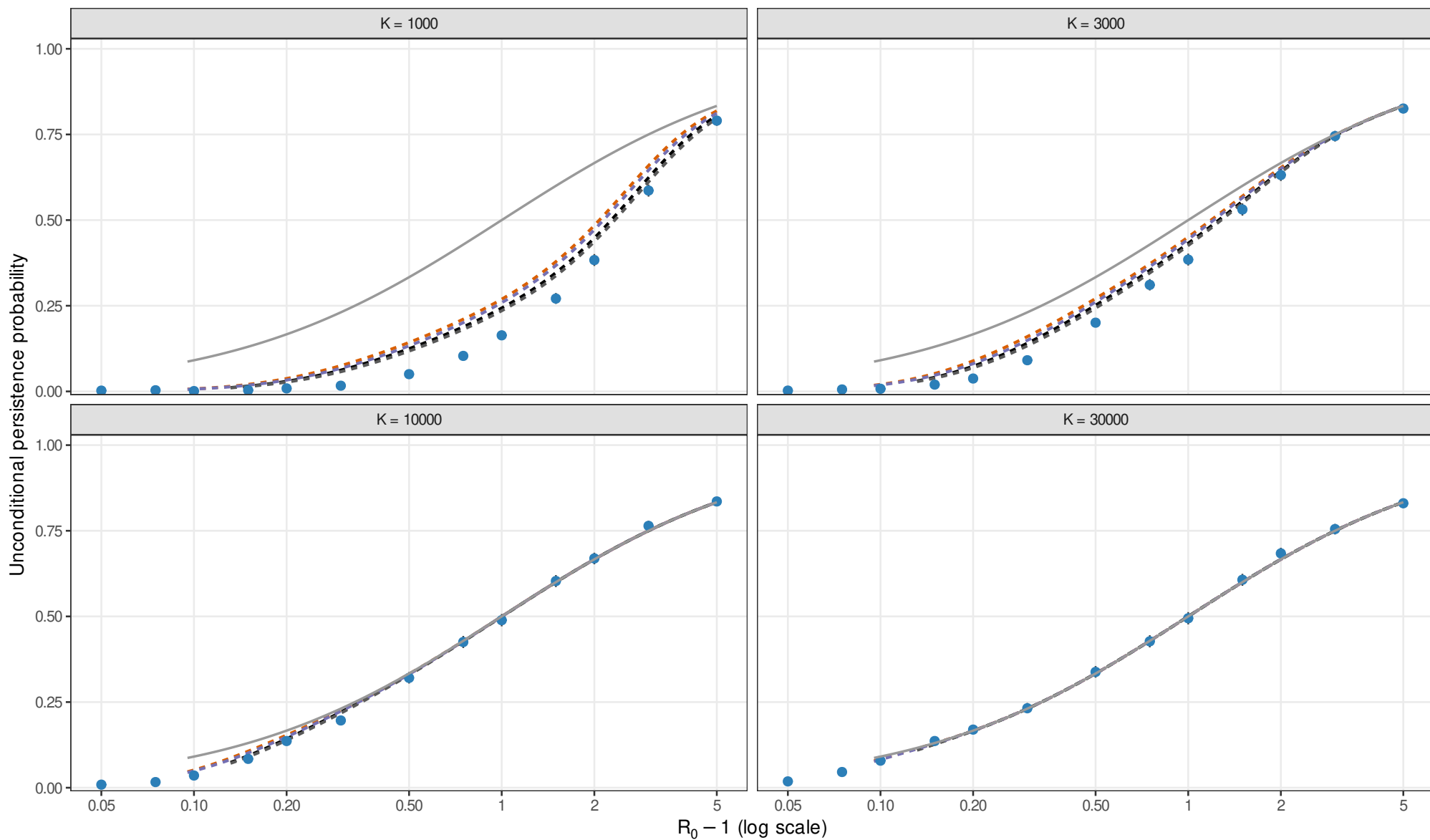
Matching height

- $y_{BL} = y_{star}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height y^* .
Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.05$, $\theta = 0$; $l_0 = 1$; matching height = y^*



Analytical method

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- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

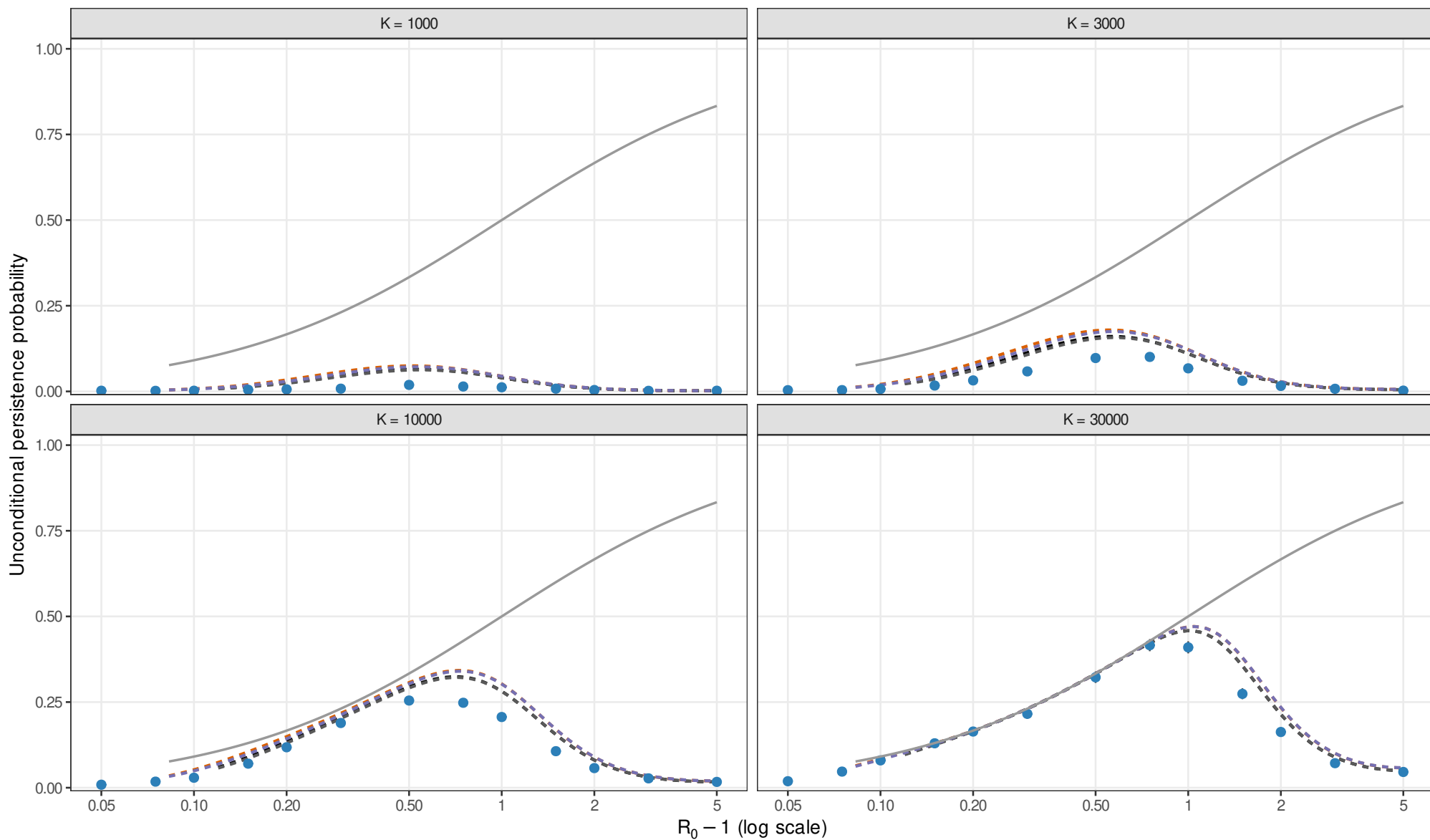
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Full stochastic validation of unconditional persistence

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Analytical method

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- Laplace + second-order entry
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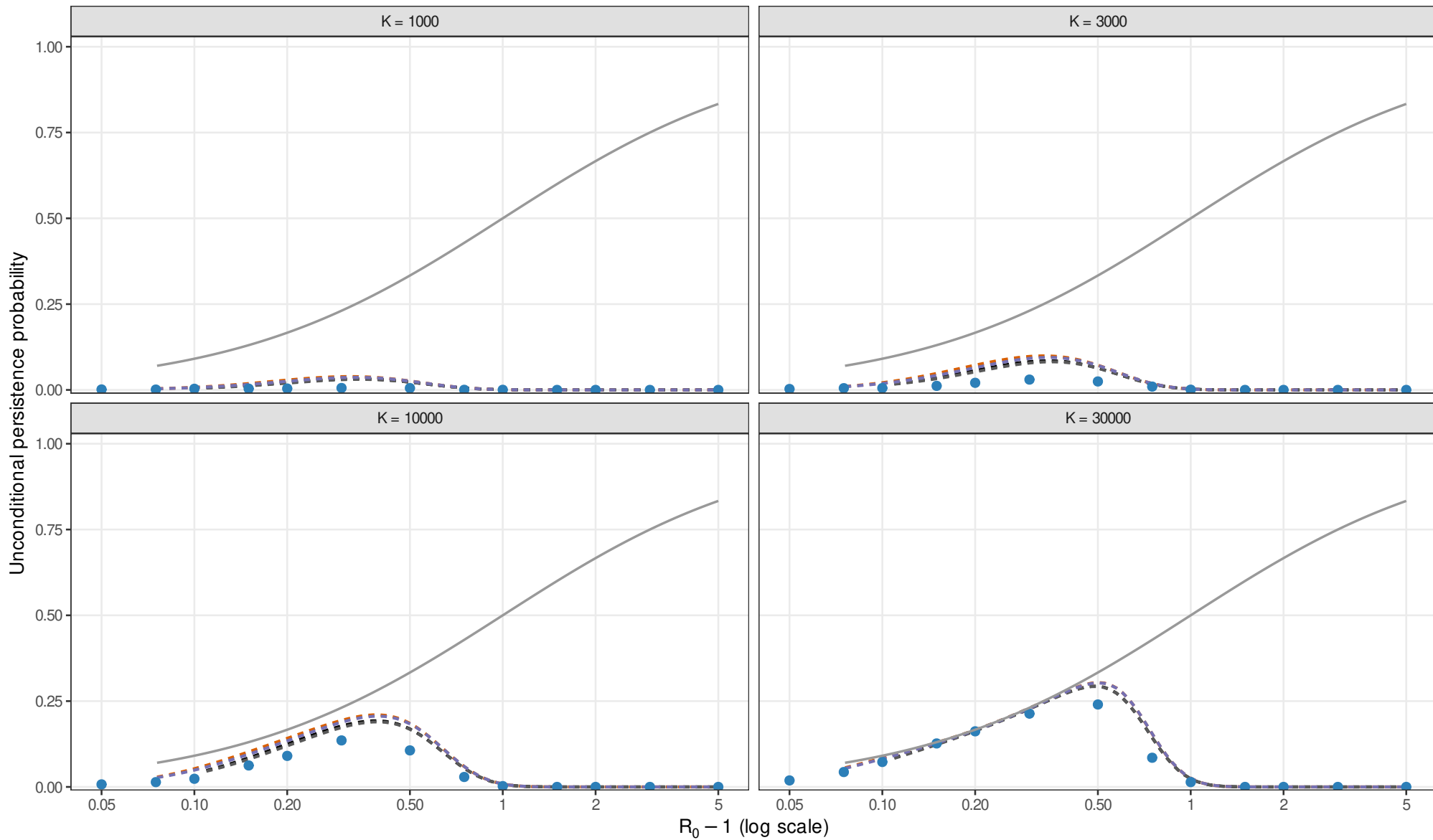
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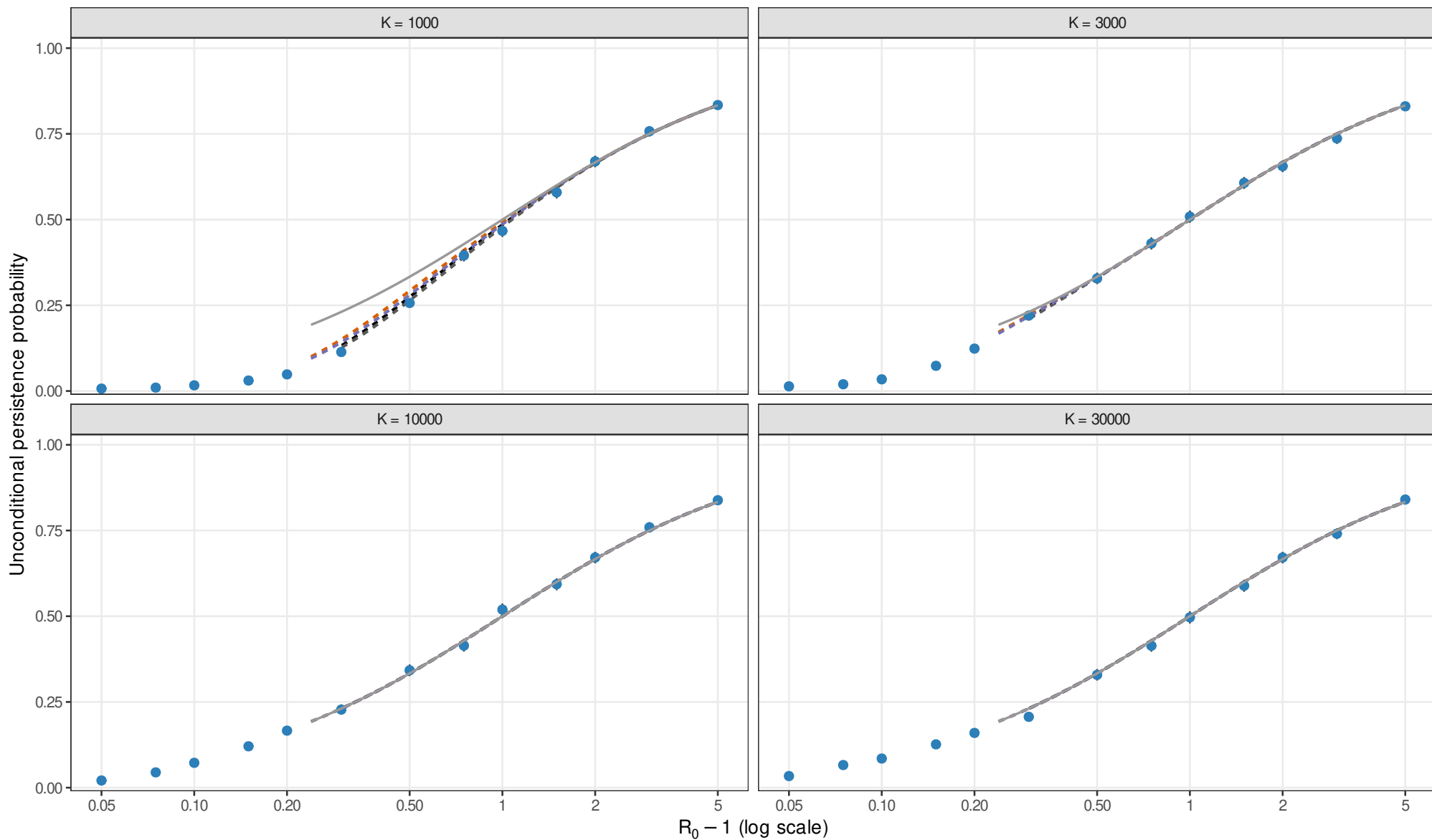
Matching height

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- Early-establishment reference

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Full stochastic validation of unconditional persistence

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Analytical method

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- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

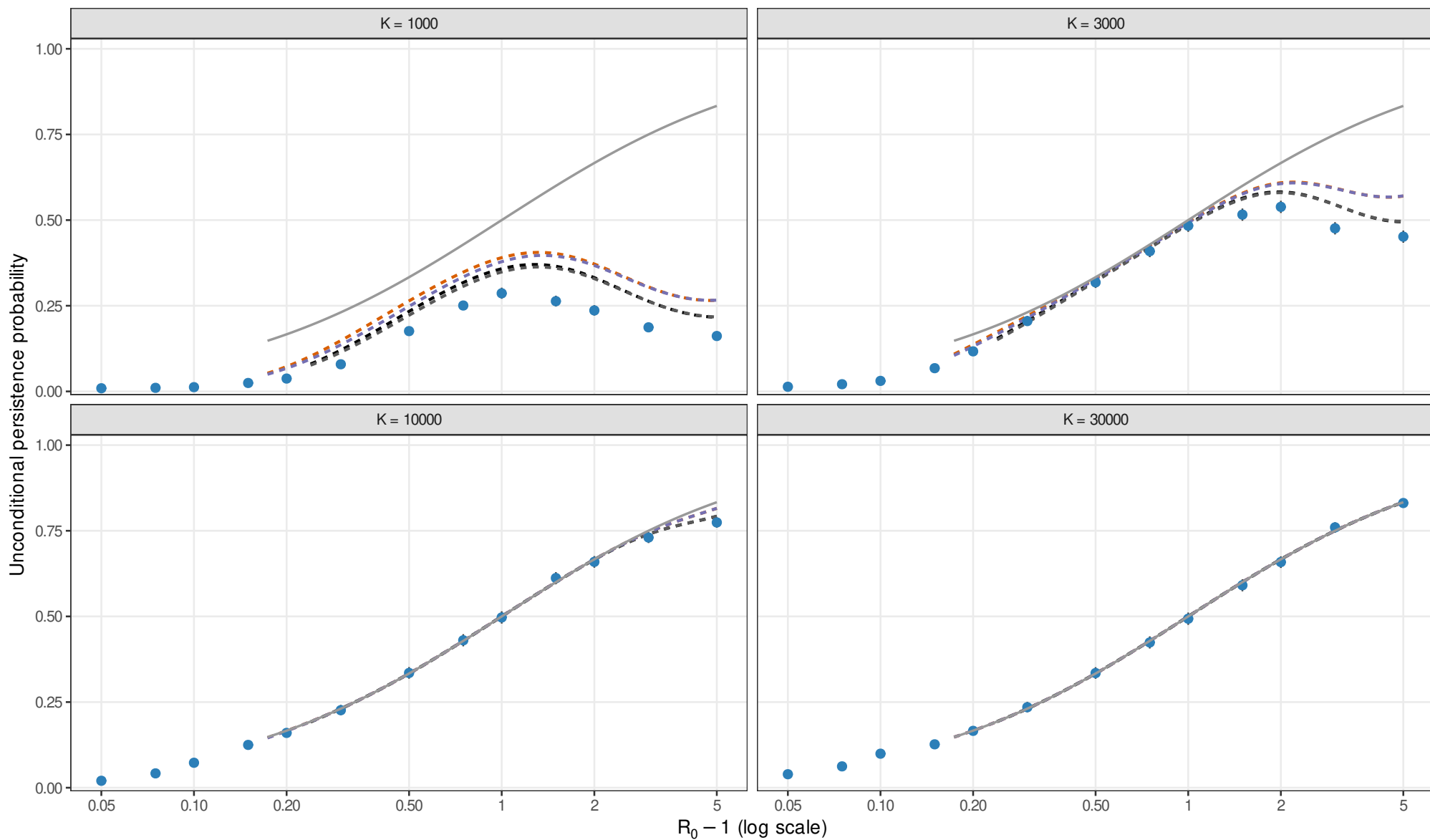
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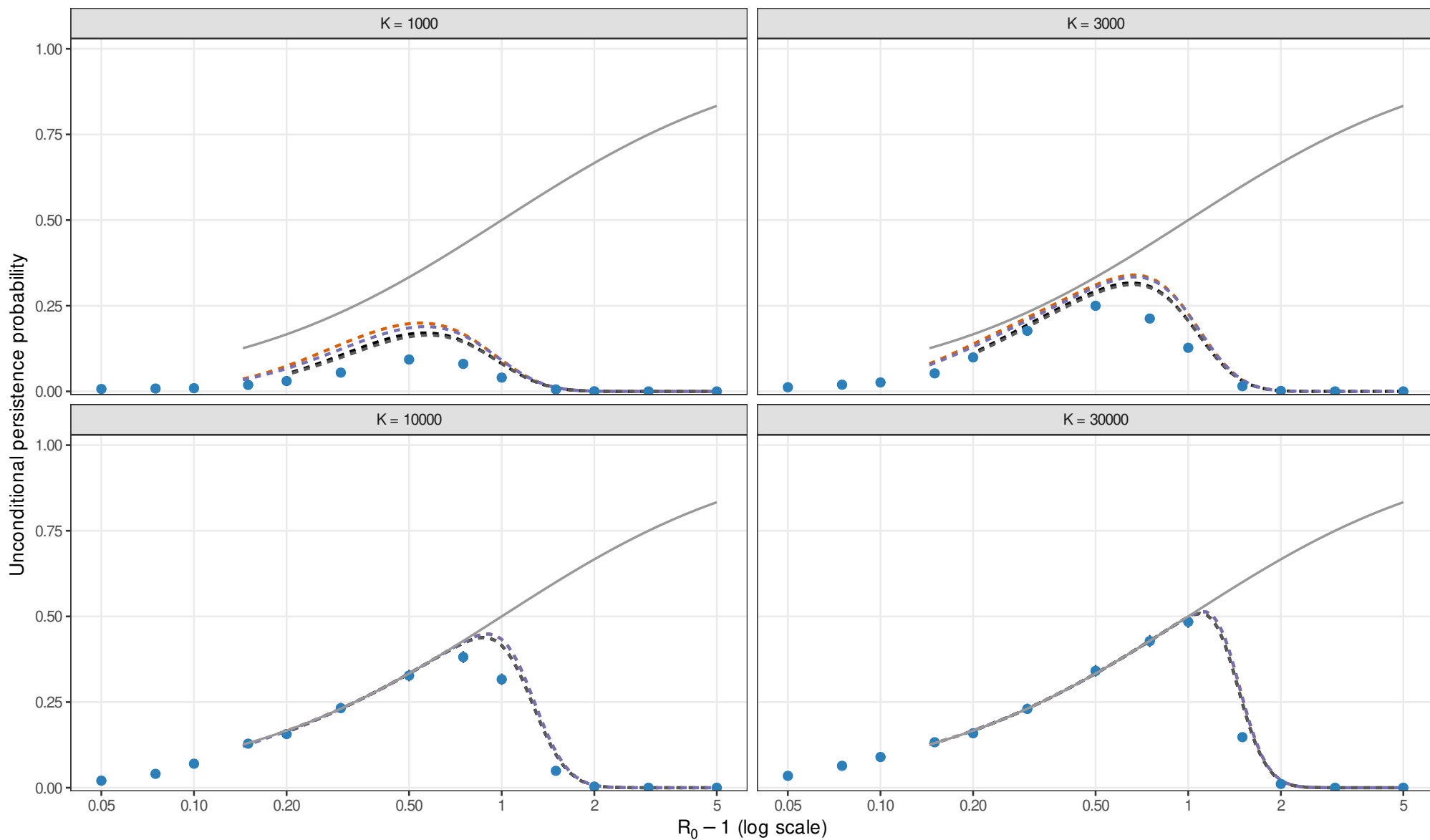
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